

Laplace Transformation:

Let f be a func defined for $t \geq 0$

Then the integral

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

is said to be the Laplace transformation^o of f provided it converges (ans is finite)

evaluate $\mathcal{L}\{1\}$

$$\begin{aligned} \underline{\text{Sol:}} &= \int_0^{\infty} e^{-st} 1 dt = \left. \frac{-e^{-st}}{-s} \right|_0^{\infty} = \boxed{\frac{1}{s}} \\ &= -\frac{1}{s} (e^{-\infty} - e^0) = -\frac{1}{s} (-1) \end{aligned}$$

Q $\mathcal{L}\{t\}$ (By Parts)

$$\begin{aligned} \underline{\text{Sol:}} &= \int_0^{\infty} e^{-st} t dt = \left. t \frac{e^{-st}}{-s} \right|_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} 1 dt \\ &= 0 + \frac{1}{s} \int_0^{\infty} e^{-st} dt = \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2} \quad s > 0 \end{aligned}$$

$$\begin{aligned} \left\{ \begin{aligned} \therefore \left. t \frac{e^{-st}}{-s} \right|_0^{\infty} &= \lim_{t \rightarrow \infty} \frac{te^{-st}}{-s} - \lim_{t \rightarrow 0} \frac{te^{-st}}{-s} \\ &= \lim_{t \rightarrow \infty} \frac{t}{-se^{st}} - 0 \\ &\text{L'Hopital } \frac{\infty}{\infty} \text{ take derivate of both Numer Denom} \\ &= \lim_{t \rightarrow \infty} \frac{1}{-s^2 e^{st}} = \frac{1}{-\infty} = 0, \quad 0 - 0 = 0 \end{aligned} \right. \end{aligned}$$

$$-\frac{1}{s}$$

evaluate: $\mathcal{L}\{t^2\}$

$$\begin{aligned}
 &= \int_0^{\infty} e^{-st} t^2 dt \\
 &= t^2 \int_0^{\infty} e^{-st} - \int_0^{\infty} \int_0^{\infty} e^{-st} \frac{d}{dt}(t^2) dt \\
 &= t^2 \cdot \frac{e^{-st}}{-s} \Big|_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} \times 2t dt \\
 &= 0 + \frac{2}{s} \int_0^{\infty} e^{-st} t dt \\
 &= 0 + \frac{2}{s} \cdot \frac{1}{s^2} \quad \text{Last question} \\
 &= \frac{2}{s^3}, \quad s > 0
 \end{aligned}$$

Q $\mathcal{L}\{t^3\} = \frac{3!}{s^4}, \quad \mathcal{L}\{t^4\} = \frac{4!}{s^5}$

$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad s > 0$ *general form*

(Gamma function)

$$\begin{aligned}
 \Gamma(n) &= \int_0^{\infty} e^{-t} t^{n-1} dt \\
 &= e^{-t} \frac{t^n}{n} \Big|_0^{\infty} - \frac{1}{n} \int_0^{\infty} t^n e^{-t} dt \\
 &= 0 + \frac{1}{n} \int_0^{\infty} e^{-t} t^n dt
 \end{aligned}$$

$$\Gamma(n) = \frac{1}{n} \Gamma(n+1)$$

$$\Gamma = n \Gamma(n)$$

$$\begin{aligned}
 \Gamma(n+1) &= \int_0^{\infty} e^{-t} t^{n+1-1} dt \\
 \Gamma(n+1) &= \int_0^{\infty} e^{-t} t^n dt
 \end{aligned}$$

$$\begin{aligned}
 &= n(n-1)\Gamma(n-1) \\
 &= n(n-1)(n-2)\Gamma(n-2) \\
 &= n(n-1)(n-2)(n-3)\Gamma(n-3) \\
 &= n(n-1)(n-2)(n-3)\dots 2 \cdot 1 \Gamma(1)
 \end{aligned}$$

$$\boxed{\Gamma(n+1) = n! \Gamma(1)}$$

using

$$\Gamma(n) = \int_0^{\infty} e^{-t} t^{n-1} dt$$

$$\Gamma(1) = \int_0^{\infty} e^{-t} t^{1-1} dt$$

$$= \left. \frac{e^{-t}}{-1} \right|_0^{\infty}$$

$$= \frac{e^{-\infty}}{-1} - \frac{e^{-0}}{-1} = 0 + 1 = \boxed{1 = \Gamma(1)} = 0!$$

$$\Gamma(n+1) = \Gamma(1) \cdot n! = n!$$

$$\because \Gamma(1) = 0! = 1$$

$$\Gamma(0+1) = 0!$$

evaluate: $\mathcal{L}\{t^n\}$

$$= \int_0^{\infty} e^{-st} t^n dt$$

$$\text{Put } st = u$$

$$t = u/s$$

$$dt = \frac{1}{s} du$$

$$= \int_0^{\infty} e^{-u} \left(\frac{u}{s}\right)^n \frac{1}{s} du$$

$$= \frac{1}{s^{n+1}} \int_0^{\infty} e^{-u} u^n du$$

$$= \frac{1}{s^{n+1}} \Gamma(n+1) = \frac{n!}{s^{n+1}} \cdot (s > 0)$$

A5Q2 $\mathcal{L}\{\cos 2t\}$

$$\frac{e^\infty}{-(s-a)} \quad \frac{1}{e^\infty} \quad \frac{1}{\infty} = 0$$

$$(e^0 = 1, e^\infty = \infty)$$

Ex: Find the Laplace transform of $f(t) = e^{at}$

$$\begin{aligned} \mathcal{L}\{e^{at}\} &= \int_0^\infty e^{-st} \cdot e^{at} dt = \int_0^\infty e^{-(s-a)t} dt \\ &= \frac{e^{-(s-a)t}}{-(s-a)} \Big|_0^\infty = 0 - \frac{1}{-(s-a)} = \frac{1}{s-a} \quad (s > a) \end{aligned}$$

Show that $\lim_{t \rightarrow \infty} \frac{e^{-(s-a)t}}{-(s-a)} = 0$

$$= \frac{e^{-\infty}}{-(s-a)} = \frac{1}{-(s-a) e^\infty} = \frac{1}{\infty} = 0$$

Q: Laplace Transformation of $\sin 2t = f(t)$

$$\mathcal{L}\{\sin 2t\} = \int_0^\infty e^{-st} \sin 2t dt$$

$$= \sin 2t \cdot \frac{e^{-st}}{s} \Big|_0^\infty - \int_0^\infty \int_0^\infty e^{-st} dt \frac{d}{dx} \sin 2t dt$$

$$= -\frac{1}{s} \left[\sin 2t \cdot e^{-st} \right]_0^\infty + \frac{1}{s} \int_0^\infty e^{-st} \cos 2t dt$$

$$= -\frac{1}{s} \left[\frac{\sin 2\infty}{e^\infty} - \frac{\sin 2(0)}{e^0} \right] + \frac{2}{s} \int_0^\infty e^{-st} \cos 2t \cdot 2 dt$$

$$= \frac{2}{s} \left[\cos 2t \cdot \frac{e^{-st}}{-s} \right]_0^\infty - \frac{2}{s} \int_0^\infty e^{-st} \sin 2t dt$$

$$= \frac{2}{s} \left[-\frac{1}{s} \left(\frac{\cos 2t}{e^\infty} - \frac{\cos 0}{e^0} \right) - \frac{2}{s} \int_0^\infty e^{-st} \sin 2t dt \right]$$

$$\frac{2}{s} \left(\frac{1}{s} - \frac{2}{s} \mathcal{L}\{\sin 2t\} \right) = \frac{2}{s^2} - \frac{4}{s^2} \mathcal{L}\{\sin 2t\}$$

$$A5Q3 = \mathcal{L}\{\cos kt\}$$

$$A5Q4 = \mathcal{L}\{\cosh kt\}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$1 + \frac{4}{s^2} = \frac{s^2 + 4}{s^2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\mathcal{L}\{\sin 2t\} = \frac{2}{s^2} - \frac{4}{s^2} \mathcal{L}\{\sin 2t\}$$

$$\mathcal{L}\{\sin 2t\} \left(1 + \frac{4}{s^2}\right) = \frac{2}{s^2}$$

$$\mathcal{L}\{\sin 2t\} = \frac{2}{s^2 + 4}$$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$\begin{aligned} \underline{\underline{Q}} \quad \mathcal{L}\{\sin kt\} &= \mathcal{L}\left\{ \frac{e^{ikt} - e^{-ikt}}{2i} \right\} \\ &= \frac{1}{2i} \left[\mathcal{L}\{e^{ikt}\} - \mathcal{L}\{e^{-ikt}\} \right] = \frac{1}{2i} \left[\frac{1}{s-ik} - \frac{1}{s+ik} \right] \\ &= \frac{1}{2i} \frac{2ik}{s^2 + k^2} \quad (i^2 = -1) \quad \frac{s+ik - s+ik}{2ik} \\ &= \frac{k}{s^2 + k^2} \end{aligned}$$

$$\begin{aligned} \underline{\underline{Q}} \quad \mathcal{L}\{\sinh kt\} &= \frac{1}{2} \left[\mathcal{L}\{e^{kt}\} - \mathcal{L}\{e^{-kt}\} \right] \\ &= \frac{1}{2} \left(\frac{1}{s-k} - \frac{1}{s+k} \right) \quad \because \mathcal{L}\{e^{at}\} = \frac{1}{s-a} \\ &= \frac{1}{2} \frac{s+k - s+k}{s^2 - k^2} \\ &= \frac{k}{s^2 - k^2} \end{aligned}$$

Q Compute Laplace Transformation of following functions

i) $f(t) = \cos^2 \omega t$

ii) $f(t) = \sin(\omega t - \phi)$

$$\therefore \cos^2 \theta = \frac{\cos 2\theta + 1}{2}$$

[A5 : Any 5 question 14.1
Q5(i-v) from book exercise

$$i) \mathcal{L}\{\cos^2 \omega t\} = \frac{1}{2} \mathcal{L}\{\cos 2\omega t\} + \frac{1}{2} \mathcal{L}\{1\}$$

$$= \frac{s^2 + 2\omega^2}{s(s^2 + 4\omega^2)}$$

$$ii) \mathcal{L}\{\sin(\omega t - \phi)\} = \mathcal{L}\{\sin \omega t \cos \phi\} - \mathcal{L}\{\cos \omega t \sin \phi\}$$

$$= \cos \phi \mathcal{L}\{\sin \omega t\} - \sin \phi \mathcal{L}\{\cos \omega t\}$$

$$\underline{Q} \quad \mathcal{L}\{e^{2(t+1)}\} = \mathcal{L}\{e^{2t} \cdot e^2\}$$

$$= e^2 \cdot \mathcal{L}\{e^{2t}\}$$

$$= e^2 \cdot \frac{1}{s-2}, \quad (s-2 > 0)$$

Q1 - Find Laplace Transformation of $f(t)$ where

$$f(t) = \begin{cases} 0 & 0 \leq t < 3 \\ 2 & t \geq 3 \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$= \int_0^3 e^{-st} f(t) dt + \int_3^{\infty} e^{-st} f(t) dt$$

$$= \int_0^3 e^{-st} \cdot 0 \cdot dt + \int_3^{\infty} e^{-st} \cdot 2 dt$$

$$= 0 + 2 \int_3^{\infty} e^{-st} dt$$

$$= \frac{2}{-s} \cdot e^{-st} \Big|_3^{\infty} = \frac{2}{-s} \left(\frac{1}{\infty} - \frac{1}{e^{-3s}} \right) = \frac{-2}{s} (0 - e^{-3s}) = \frac{2}{s} e^{-3s}$$